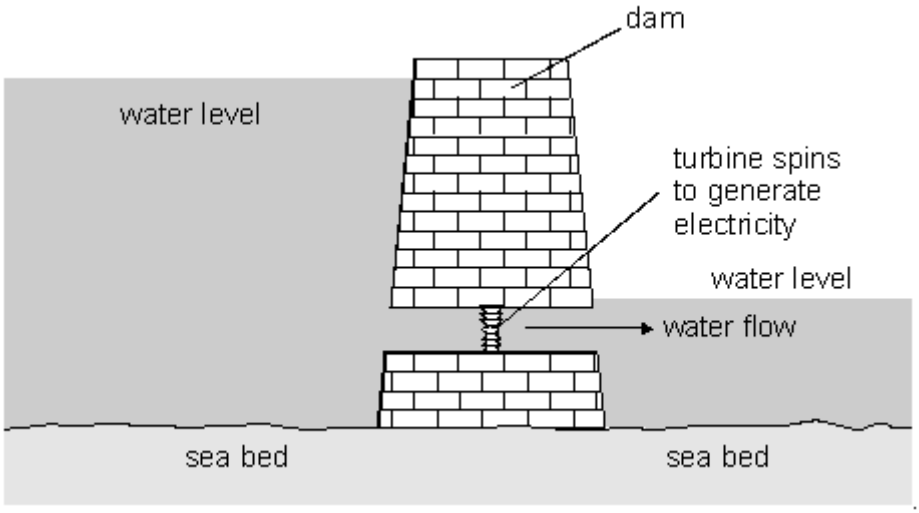
Q1. The tides can be used to generate electricity. A dam is built across a river estuary, as shown below.



(a) The water is higher on one side of the dam than on the other. As the water begins to flow through the dam it turns a turbine. The turbine generates electricity.
Describe the useful energy changes which take place in this process.

.....

.....

.....

.....

2 marks

(b) Explain why tides are classified as a renewable energy source.

.....

.....

1 mark

(c) Give **one** way, **other** than from the tides, of generating electricity by using the sea.

.....

1 mark

(d) Apart from cost, give **one** advantage and **one** disadvantage of an oil-fired power station compared with a tidal power station.

advantage .....

.....

disadvantage .....

.....

2 marks
Maximum 6 marks

Mark schemes

Q1.

(a) **The first marking point is for the transfer of energy from water to turbine.**
The second marking point is for the transfer of energy from turbine to generator.
The third marking point is for the transfer of energy away from the generator.

any **two** from

- potential energy in the water to kinetic energy in the turbine
  accept 'P.E. to K.E.'
  accept 'transferred from the water to the turbine'
  accept 'K.E. in the water to K.E. in the turbine'
  accept 'P.E. in the water to K.E. in the water'

- kinetic energy in the turbine to kinetic energy in the generator
  accept 'transferred from the turbine to the generator'

- kinetic energy in the generator to electrical energy in the circuit
  accept 'KE. to electrical energy'
  accept 'from the generator to the circuit'
  accept 'transferred from the generator by electricity'
  accept 'KE. in the turbine to electrical energy in the circuit'
  accept 'potential energy in the water to electrical energy in the circuit' for both marks
  accept 'P.E. to electrical energy'
  ***or** 'from the water to the circuit' for one mark*

2

(b) any **one** from

- because the Moon's pull **or** gravity is always there

- because the tides **or** the water cannot run out **or** be used up
  accept 'because there are tides every day'
  ***or** 'because there is an endless supply'*

1

(c) • from wave energy **or** from the waves
  *accept 'Ocean Thermal Energy Conversion' **or** 'OTEC'*
  *do **not** accept 'hydro-electric power'*

1

(d) it is easier to control or it can be turned on when it is needed
  accept 'the tides only give power at certain times'
  ***or** 'you can build an oil-fired power station anywhere'*
  ***or** 'it is smaller`*

any **one** from

- oil is non-renewable
  accept 'oil will run out'

- it causes pollution
  accept 'it gives out greenhouse gases'

or *'it can cause oil spills'*

1

[6]